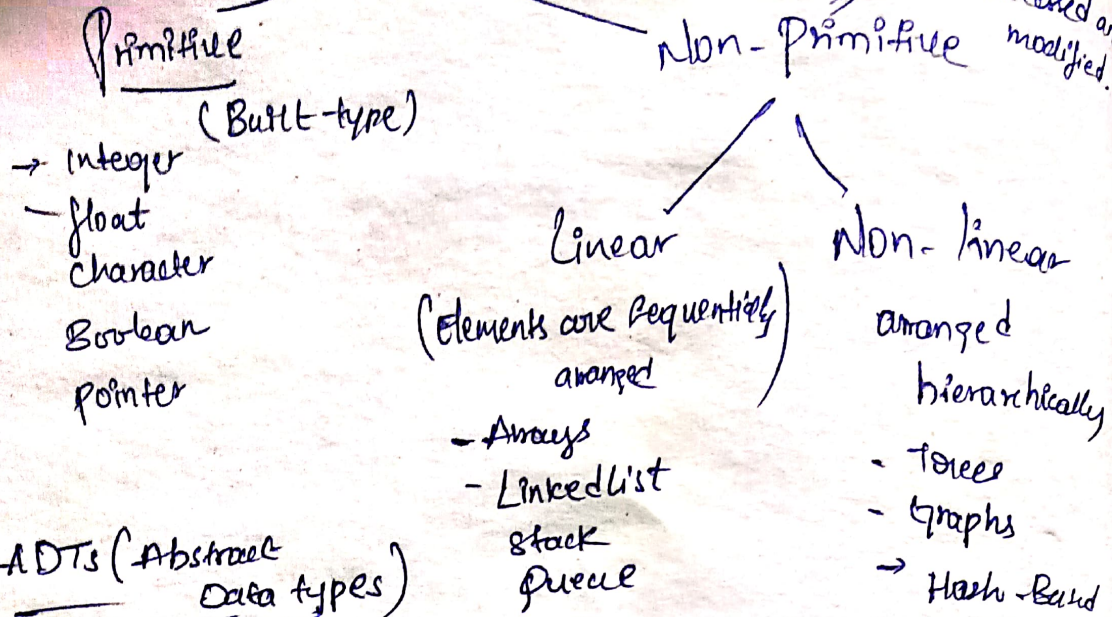


Data Structures (A way of organising & storing and managing data in memory) can be accessed or modified.



ADTs (Abstract Data types)

These defines behaviour rather than implementation

- List
- Queue
- Stack
- Set
- Map
- Priority Queue

Primitive → A basic built-in data type that store single value

Non-Primitive → A data structure that stores and organize multiple values

ADT : A logical model that defines what operations can be performed on data structure without specifying how they are implemented

- Basics → Primitive
- Built-in functions
- Time Complexity (10)
- Space Complexity (8)
- Recursion (3)
- Bit manipulation (1)
- Java collection framework (2)
- Algorithms.

$$\begin{aligned}
 n \log n &= n \times 2k \\
 &= (n \times 2) \\
 &= 5 \times 2 = 10
 \end{aligned}$$

↓
Happy
= char(2) → 6

$(n > 0) \text{ and } (n \> (n-1)) \Rightarrow 0$
 $2 \rightarrow 0010$
 $4 - 0100$
 $8 \rightarrow 1000$
 $16 \rightarrow 0111$
 $32 \rightarrow 0000$
 $64 \rightarrow 0110$
 $128 \rightarrow 0110$
 $256 \rightarrow 0110$
 $512 \rightarrow 0110$
 $1024 \rightarrow 0110$
 $2048 \rightarrow 0110$
 $4096 \rightarrow 0110$
 $8192 \rightarrow 0110$
 $16384 \rightarrow 0110$
 $32768 \rightarrow 0110$
 $65536 \rightarrow 0110$
 $131072 \rightarrow 0110$
 $262144 \rightarrow 0110$
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 $2097152 \rightarrow 0110$
 $4194304 \rightarrow 0110$
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 $33554432 \rightarrow 0110$
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 2208

Collection Types

- List
- Set
- Queue
- Stack

MAP types

- HashMap (No order)
- LinkedHashMap (Insertion order)
- TreeMap (Sorted by key)

SET

- HashSet (No order)
- LinkedHashSet (Insertion order)
- TreeSet (Sorted order)
- (No Duplicates)

List (Ordered + Indexed)

Allows duplicates

- ArrayList (Ordered) (Not sorted)
- LinkedList " " " "

Hash → fastest but ordered

linked → maintains order

Tree → sorted automatically

Bit Manipulation

Left shift ($\ll$)

$$n \ll k = n \times 2^k$$

Right shift ($\gg$)

$$n \gg k = n / 2^k$$

① Check kth Bit → check "1"

$$(n \& (1 \ll k)) \neq 0$$

② Clear kth Bit → "0"

$$n \& \sim (1 \ll k)$$

③ Set kth Bit → change to "1"

$$n | (1 \ll k)$$

④ Toggle kth bit → 1 → 0 (specify bit)
0 → 1

$$n \wedge (1 \ll k)$$

⑤ Invert All Bits → all bit reverses
1 → 0 (Position changes)
0 → 1

$$\sim n$$

$$n \& (1 \ll k)$$

→ anything

$$0111 = 0000101$$

$$1010000$$

⑥ Power of 2

$$(n > 0) \& (n \& (n-1)) == 0$$

$$n=8$$

$$1 \ll 3$$

8-2

16	8	4	2	1
0	0	0	0	0

⑦ Integer to Binary String()

⑧ Integer.bitCount()

⑨ Integer.highestOneBit()

0 → check
 1 → set
 ~ → toggle
 shift → move
 $n \& (n-1)$ → Remove rightmost set bit
 $1 \ll k$ → create mask

Built-in functions

String

str.length
str.charAt()
str.substring(start)
str.substring(start, end)
↑
not included
str.indexOf
str.contains("text")
str.equals(str2)
str.equalsIgnoreCase(str2)
str.toLowerCase()
str.trim → " Hello "
str.replace → ^{removed} ("x", "x") { "Java, Python" }
str.split → str.split(",") = b[] string.
str... → array.toString(b)
str.isEmpty() → array.sort()

Collections

collections.sort()
collections.reverse()
collections.max()
collections.frequency()
collections.shuffle()
collections.sort(list, reverse=false);
→ ascending

HashMap

put()
get()
remove()
containsKey()
containsValue()
keySet
values()
entrySet()
size()
isEmpty()

Character

Character.toUpperCase()
Character.toLowerCase()
Character.isLetter()
Character.isUpperCase()

StringBuilder

append()
insert()
delete()
deleteCharAt()
reverse()
charAt()
setCharAt()
length()
toString()

Math

Math.abs()
Math.max
Math.sqrt()
Math.pow(a, b)
Math.ceil() → up
Math.floor() → down
Math.random → nearest
or
Math.random

ArrayList

add()
remove()
get()
set()
size()
contains
indexOf
isEmpty()
"clear"

HashSet

add()
remove()
contains()
size()
isEmpty()

Queue

offer()
poll
peek
isEmpty()
size()

Stack

push
pop
peek
isEmpty()
search()

Recursion

Cheat codes

- * A recursion that which a recursive function solves a problem by solving a smaller version of itself.
- * A function calls itself to solve smaller versions of the same problem

Important part

- Base case (condition that stops recursion)
- Recursive call (function calls itself with a smaller problem)

Execution

fun(3)
└─ fun(2)
 └─ fun(1)
 └─ fun(0)

$5 \times \text{fact}(4)$
 $5 \times 4 \times \text{fact}(3)$
 $\vdots$

flow
↓ call
 call
 Base case
 ↓
 return
 ↓
 return

- factorial
 return $n \times \text{fact}(n-1);$
- Sum of numbers
 return $n + \text{sum}(n-1);$
- Fibonacci
 return $\text{fib}(n-1) + \text{fib}(n-2)$
- Power
 return $a \times \text{power}(a, b-1)$
- Reverse array
 reverse(arr, start+1, end-1)
- Palindrome
 return palindrome(s, l+1, j-1);